Q1) the output is:
hey from message1

There is only one possible output because there is only one bean notation

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Q2) the output is:
hey from message1
hey from message2

There is only one possible output because getMessage2() depends on the value returned by getMessage1() for that reason spring boot must initialize getMessage1() first then getMessage2()

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Q3) the output is:
hey from message1
hey from message3
hey from message2

This output is possible because getMessage1() is independent and can be initialized first after that spring initializes getMessage3() and
then getMessage2() because getMessage2() depends on getMessage3()

or
hey from message3
hey from message1
hey from message2

This output is also possible because spring may initialize getMessage3() first
since getMessage1() is independent it can be initialized after getMessage3() and before getMessage2() finally getMessage2() is initialized after getMessage3() because it depends on it.

or
hey from message3
hey from message2
hey from message1

This output is possible because spring may initialize the dependent chain first It starts with getMessage3() then getMessage2() because it depends on getMessage3() after that getMessage1() can be initialized because it is independent.

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Q4) the output is:
hey from message3
hey from message2
hey from message1
hey from Main controller

This output is possible because Spring may complete the second dependency chain first by initializing message3 and then message2(). After that, it initializes message1, and finally MainController, which depends on message1.

or
hey from message1
hey from Main controller
hey from message3
hey from message2

This output is possible because message1 must be initialized before MainController, so spring creates that chain first then it initializes message3 followed by message2 because message2 depends on message3().

or
hey from message3
hey from message1
hey from message2
hey from Main controller

This output is possible because Spring may initialize message3 first, then message1. After that, message2() can be initialized because message3 is already available. Finally, MainController is initialized after message1.

or
hey from message3
hey from message1
hey from Main controller
hey from message2

This output is possible because Spring may start with message3(). Then it can initialize message1(). Since MainControllerdepends only on message1, it can be created next. Finally, message2() is initialized after message3().

or
hey from message1
hey from message3
hey from message2
hey from Main controller

This output is possible because after initializing message1, Spring may initialize the second chain completely by creating message3 first and then message2().
Since MainController only depends on message1, it can be initialized after that.

or
hey from message1
hey from message3
hey from Main controller
hey from message2

This output is possible because Spring initializes message1 first, then it may initialize message3. After that, MainController can be initialized because its dependency message1 is already available. Finally, message2 is initialized after message3().

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Q5) the output is:
hey from message3
hey from message2
hey from Main controller
hey from message1

There is only one possible output because all methods form a single dependency chain:

- getMessage2() depends on getMessage3()
- MainController depends on getMessage2()
- getMessage1() depends on MainController